Beyond the Model: How Harness Design Affects LLM Performance on the 2026 International Mathematical Olympiad

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Abstract

Two frontier systems, GPT-5.6 Sol and Claude Fable 5, solved the 2026 International Mathematical Olympiad. Each scored $42/42$ on six contest problems released after their training cutoff and attempted with no internet access, and did so through every harness that could return complete output. For the sub-frontier models the score moves sharply with the harness, while for the frontier models it barely moves at all. Claude Sonnet 5 scores $14/42$ through the web interface, $21/42$ through the provider's Claude Code agent, and $35/42$ through the multi-agent harness AutoFyn, crossing the 29-point gold cutoff at the 2026 contest only in the last [International Mathematical Olympiad, 2026]. Claude Opus 4.8 spans the same range, and the open-weight GLM 5.2 goes from $14/42$ to $28/42$, one point short of gold. The two frontier systems, by contrast, stay at or near a perfect score across every harness they run. To trust these numbers, we graded every proof with a frontier model other than its author, under a standard that awards zero for any unproved load-bearing step, then had a panel of three past IMO medalists led by a gold medalist review the grades, and archived every run's full process trail publicly.[1]

1 Introduction

The International Mathematical Olympiad is among the hardest tests of human mathematical reasoning. Six hundred of the strongest students in the world sit six problems over two days, and fewer than one percent reach a perfect $42/42$ [International Mathematical Olympiad, 2025]. AI reached only the silver threshold there in 2024 [Hubert et al., 2025], and by 2025 general-purpose models were reported at gold level in natural language [OpenAI, 2025, Google DeepMind, 2025, Huang and Yang, 2025]. Gold is now the year-old bar; the open target is a perfect score. Whether the frontier can produce a flawless paper on a fresh olympiad, under grading anyone can check, remains open, as does how much of the result belongs to the model rather than the harness it runs through.

To address this, we put the current frontier on the 2026 IMO, whose six problems were released after every model's training cutoff, with web search disabled throughout. We evaluate five models, each through up to three harnesses of increasing structure, a plain web chat interface, the provider's local agent (Claude Code or Codex), and the multi-agent harness AutoFyn [Hasan et al., 2026], and grade every proof under a harsh completion standard that awards full marks only for a complete proof and zero for any unproved load-bearing step. To avoid confirmation bias, we grade each proof with a frontier model other than the one that produced it, and have a panel of three past IMO medalists led by a gold medalist review the grades. We also archive every run's full process trail publicly. Under this standard, GPT-5.6 Sol and Claude Fable 5 each produce complete, rigorous proofs of all six

[1] https://github.com/SignalPilot-Labs/autofyn/tree/main/results/imo-2026

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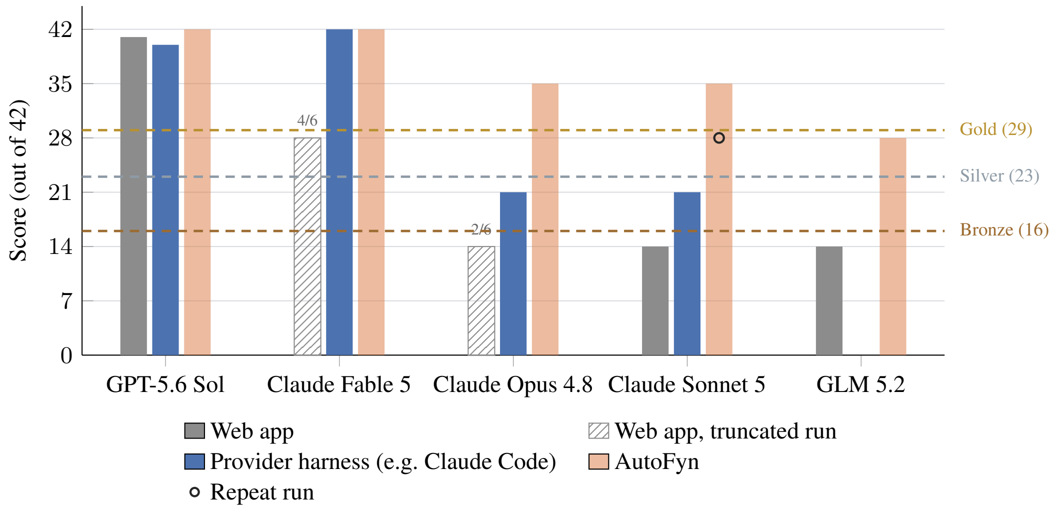

Figure 1: LLM Model and Harness Performance at IMO 2026: Audited score of each model under each harness, out of 42. Dashed lines mark the medal cutoffs at the 2026 contest; no causal ranking of harnesses is implied (Section 3.3). Hatched bars mark web runs truncated by the interface's output-token limit, which returned no gradable output on some problems after five retries; they are graded on 2 of 6 (Claude Opus 4.8) and 4 of 6 (Claude Fable 5) problems (Section 5.1). GLM 5.2 has no provider-agent run. Solid dots mark repeat runs of a cell, plotted over the relevant harness bar; the reported cell is always the archived run, and a dot indicates run-to-run spread (Section 6). The dot on Claude Sonnet 5 is a second AutoFyn run, which scored $28/42$ against the archived run's $35/42$.

problems and score $42/42$, while Claude Opus 4.8 and Claude Sonnet 5 each reach $35/42$ through AutoFyn, above the 29-point gold-medal cutoff at the 2026 contest [International Mathematical Olympiad, 2026], and the open-weight GLM 5.2 reaches $28/42$.

Our central finding is that the harness barely moves the frontier models but is decisive below it. The frontier models solve all six problems in their own provider harness, while for the sub-frontier models the harness determines which medal, if any, a run earns. Against the 2026 cutoffs of 16, 23, and 29 points for bronze, silver, and gold [International Mathematical Olympiad, 2026], Claude Sonnet 5 finishes below any medal through the web interface ($14/42$), at bronze through Claude Code ($21/42$), and at gold through AutoFyn ($35/42$), and Claude Opus 4.8 climbs from bronze to gold the same way. The open-weight GLM 5.2, run through the web and AutoFyn only, rises from a below-medal $14/42$ to $28/42$, one point short of the gold cutoff. The frontier models, by contrast, move little across harnesses, GPT-5.6 Sol spanning two points over its three, so the gap that decides a medal one tier down all but vanishes at the top. Figure 1 shows the effect across every model we ran.

The gains under the multi-agent harness fall on two problems whose load-bearing step matches something the harness provides. Problem 2 (geometry) closes on a long exact calculation backed by a symbolic certificate, and the sub-frontier certificates that earn full marks are the ones AutoFyn retains for the auditor to re-run. Problem 6 (number theory) has a past analogue, IMO Shortlist 2013 N5, which the harness's explorer surfaces from a corpus of prior problems; the sub-frontier models adapt it and solve Problem 6 only under AutoFyn, while GPT-5.6 Sol reaches the same result on its own in all harnesses. Problem 3 (combinatorics) matches neither: its load-bearing step has no analogue to retrieve and no certificate to check, and no harness closes the gap. The harness supplies verification and retrieval; it does not supply an argument the model cannot find.

The rest of the paper follows the evidence. Section 4 presents the audited results, and Section 5 presents our analysis: where the systems fail, how they solve problems, one graded run shown in full, and what the harness adds.

2 Related Work

It is known in various domains that a fixed model's score depends on the scaffold around it. For instance, in software engineering the agent-computer interface governs task success [Jimenez et al., 2024, Yang et al., 2024, Shinn et al., 2023], and in mathematics a verification-and-refinement pipeline lifts fixed models from roughly 20–40% to gold on IMO 2025 [Huang and Yang, 2025], yet the harness is rarely treated as an explicit evaluation axis. The heaviest harness in our matrix, AutoFyn [Hasan et al., 2026], sits in this line of work, an open multi-agent system that assigns separate subagents to explore approaches, retrieve analogous past problems, prove, and review before a proof is finalized.

As frontier models get smarter, measuring their capability also becomes harder. As of 2026, the standard math benchmarks are saturated and contamination-prone [Hendrycks et al., 2021, Cobbe et al., 2021, Mirzadeh et al., 2024], which has driven evaluation onto fresh competitions [Glazer et al., 2024, Balunović et al., 2025]. Grading full proofs by model is itself unreliable, since models are biased as judges [Zheng et al., 2023] and miss hard-to-detect proof errors even when the surface reads well [Dekoninck et al., 2025].

Against this, our unit of study is the model-and-harness system rather than the model, our contest is fresh and post-training-cutoff-date, and our grades come from an independent cross-model and human-medalist audit rather than a vendor or a self-assessment.

3 Setup

Each result we report is one cell of a model-by-harness matrix: a single archived run of one model, in one harness, over all six 2026 problems, graded under a fixed standard. This section defines that cell, the grading and audit it passes through, and which comparisons across cells are controlled. Per-cell results appear in Section 4.

3.1 The Evaluation Matrix

We evaluate five models, Claude Opus 4.8, GPT-5.6 Sol, Claude Fable 5, Claude Sonnet 5, and the open-weight GLM 5.2, across three harnesses, an evaluation matrix with 14 of 15 cells run: GLM 5.2 runs through the web and AutoFyn harnesses only, with no provider-agent run. The first harness is the web application with web search disabled. The second is the provider harness, the provider's own local agent, Claude Code (v2.1.209) for the Claude models and the Codex CLI (v0.142.5) for GPT-5.6 Sol, with every run in a given agent executed on that fixed version. The third is AutoFyn, a strong open multi-agent harness that explores several approaches per problem, searches a curated corpus of past olympiad problems for analogous techniques, extracts reusable lemmas, and runs an internal reviewer before finalizing a proof [Hasan et al., 2026]. Every cell is one archived, audited run over all six problems, with no averaging across attempts. Additionally, the problems are beyond every model's training cutoff and no 2026 solution material was supplied to a run or retrievable through its declared environment. So, freshness is scoped to the declared environment and holds equally across all harnesses.

3.2 Grading and Independent Audit

We score all cells under one strict completion-based standard. We give a complete proof 7, and a proof missing a load-bearing component, such as an entire upper or lower bound, 0, with no partial credit for a sound structure around the gap. We make one exception: when a proof is complete in substance, with no load-bearing step missing, but carries a typo or an unproved minor step whose repair is obvious, we deduct one or two points after panelist discussion. We do not score a problem that produced no gradable output, record it as *n/g*, and exclude it from the denominator.

We adopt this near-binary standard rather than graduated partial credit because partial credit requires the grader to assign a fair value to an incomplete argument. In practice, this is difficult to determine consistently and increases the burden of reaching agreement between AI and human graders. Restricting the decision to the completeness and soundness of the proof makes the grade reproducible and admits a unanimous panel verdict.

To prevent confirmation bias in grading, no model graded its own run. Each archived run is audited by a distinct frontier model under a fixed reviewer prompt, with Claude runs audited by GPT-5.6, GPT-5.6 runs audited by Claude Fable 5, and GLM 5.2 runs audited by Claude Fable 5, except the GPT-5.6 Sol web run, which was audited by Gemini. This audit produces the initial assessment, which the authors adopt. The model audits are then reviewed by a panel of three past IMO medalists, one a gold medalist who leads the panel. The scores reported here are therefore the authors' assessment together with this medalist review, and are not official IMO jury grades.

The panelist review was indeed necessary. In some cells the panel revised the model audit, for instance lowering GPT-5.6 Sol's web Problem 2 from 7 to 6 and Claude Sonnet 5's AutoFyn Problem 3 from 7 to 0 once a load-bearing step in the lower bound was found to be missing.[2]

3.3 On Comparing Harnesses

Each cell runs a harness as shipped, at its default configuration, the operating point a user gets out of the box. We impose only two uniform limits on top of each harness's own defaults: a wall-clock budget of 20 hours per problem, and, when a system fails to return any gradable output on its first attempt, up to five retries on the same problem before we record the cell as ungradable. The retry rule matters mainly for the web interfaces, where a single attempt can terminate early on a message-length limit rather than on the mathematics (Section 5.1). A run ends when the system submits its final write-up, exhausts its retries, or reaches the 20-hour budget, so each cell answers one question, whether that deployment produces complete proofs when left to work. Limits internal to a deployment, such as the web application's message-length cap, are part of what is being measured and are recorded as such (Section 5.1).

The two directions of the matrix carry different evidential weight. Comparing models within a single harness holds the harness, its version and defaults, the problems, and the grading fixed, so these comparisons are controlled; the five AutoFyn rows of Table 1 differ only in the model. Comparing harnesses for a fixed model spans cells that differ on tools, orchestration, and effective budget at once, so these comparisons are descriptive; we report how outcomes vary with the harness without attributing the difference to any single ingredient, and the reported times are episode lengths rather than times to solution.

4 Results

Table 1 gives the audited scores on the six fresh problems of the 2026 IMO, under the completion-based standard of Section 3.2. Table 2 records the wall-clock process metadata for the same cells.

Two systems reach $42/42$, GPT-5.6 Sol through AutoFyn and Claude Fable 5 through both Claude Code and AutoFyn, and GPT-5.6 Sol stays within two points of perfect in its other two harnesses. Claude Opus 4.8 and Claude Sonnet 5 peak at $35/42$, both through AutoFyn. The newer and smaller Claude Sonnet 5 thus matches the previous-generation flagship, identical $21/42$ totals through Claude Code and identical $35/42$ through AutoFyn with the same single missing problem. The open-weight GLM 5.2, run through the web and AutoFyn only, climbs from $14/42$ to $28/42$, one point below the 29-point gold cutoff; its web total ties Claude Sonnet 5's, two model families reaching the same $14/42$ through their web interfaces and both gaining 14 or more points through AutoFyn. Within a fixed model the total moves with the harness, by 2 points across GPT-5.6 Sol's three harnesses, by 21 across Claude Sonnet 5's, and by 14 across GLM 5.2's two, taken up in Section 5.4. The n/g entries in the two Claude web runs are the message-length limit of the interface, described in Section 5.1, and the zeros are analyzed in Section 5.1.

5 Analysis

We analyze the audited runs along three axes: how the systems fail, how they solve, and what the harness contributes, with one graded run shown in full.

[2]The panel likewise adjudicated GLM 5.2's AutoFyn Problem 2 to 0 from an initial model assessment of $2/7$, judging the remaining gaps load-bearing under the completion standard.

Table 1: Per-problem scores (out of 7) by model and harness, under the completion-based standard and audit protocol of Section 3.2; authors' assessment, not official IMO grades. "n/g" marks a problem with no gradable output, excluded from the denominator, so a total x/y with $y < 42$ counts only gradable problems; the dagger (†) marks the Claude Opus 4.8 web run, gradable on two of six problems. GLM 5.2 was run through the web and AutoFyn harnesses only.

Model	Harness	P1	P2	P3	P4	P5	P6	Total
Claude Opus 4.8	Web (claude.ai)	n/g	n/g	n/g	7	7	n/g	14/14†
Claude Opus 4.8	Claude Code	7	0	0	7	7	0	21/42
Claude Opus 4.8	AutoFyn	7	7	0	7	7	7	35/42
GPT-5.6 Sol	Web (chatgpt.com)	7	6	7	7	7	7	41/42
GPT-5.6 Sol	Codex	7	5	7	7	7	7	40/42
GPT-5.6 Sol	AutoFyn	7	7	7	7	7	7	42/42
Claude Fable 5	Web (claude.ai)	7	7	n/g	7	7	n/g	28/28
Claude Fable 5	Claude Code	7	7	7	7	7	7	42/42
Claude Fable 5	AutoFyn	7	7	7	7	7	7	42/42
Claude Sonnet 5	Web (claude.ai)	7	0	0	0	7	0	14/42
Claude Sonnet 5	Claude Code	7	7	0	0	7	0	21/42
Claude Sonnet 5	AutoFyn	7	7	0	7	7	7	35/42
GLM 5.2	Web (chat.z.ai)	7	0	0	7	0	0	14/42
GLM 5.2	AutoFyn	7	0	0	7	7	7	28/42

5.1 How the Systems Fail

The frontier models, GPT-5.6 Sol and Claude Fable 5, miss no problem in the agent harnesses. GPT-5.6 Sol's one weak point is Problem 2 (geometry), whose certificate requires a long exact calculation that it completes in full only under AutoFyn, scoring $6/7$ and $5/7$ in the two lighter harnesses. The sub-frontier models fail on the hardest problems, Problems 2, 3, and 6, and Claude Sonnet 5 also on Problem 4.

On Problem 2 the step is the closing algebra. All seven $7/7$ cells close with a symbolic certificate that the auditor checked or re-ran, and every lost point sits at that same step. Claude Sonnet 5 on the web could not derive the target equality, Claude Opus 4.8 in Claude Code asserted the decisive ideal membership without a certificate, and GPT-5.6 Sol displayed a false factorization through Codex ($5/7$) and left sign and nondegeneracy conditions unproved on the web ($6/7$). The audits' shared policy is that a claim of verified algebra earns nothing without a checkable certificate.

On Problem 3 (combinatorics) every successful attempt goes through one hard step. Every model reaches the correct value and the same threshold-parity monovariant, so the answer and the high-level strategy are shared; the outcome turns entirely on one reduction. GPT-5.6 Sol closes this reduction in all three harnesses, on the web in 13 minutes, and Claude Fable 5 in both harnesses where it produced output. Claude Opus 4.8 and Claude Sonnet 5 derive the same signed-score representation but leave the reduction open in every run. The strongest failing attempt, Claude Sonnet 5 through AutoFyn, proves the value and the full upper bound over twenty hours and stalls at exactly this step. No past-problem precedent was found for the reduction, and neither added orchestration nor added time could close it.

A third class of failure is mechanical. On the web interface the Claude Opus 4.8 run hit the maximum output length on four problems and the Claude Fable 5 run on two, returning no output across five retries, so those attempts are ungradable rather than wrong. Fable 5 scores $42/42$ through both agent

Table 2: Per-run process metadata in integer minutes. Episode or wall-clock durations with no correctness conclusion. AutoFyn figures are episode length rather than time to solution (Section 3.3); its episodes run longer because several subagents explore, prove, and review in sequence, not because one model thinks longer.

Model	Harness	P1	P2	P3	P4	P5	P6	Total
Claude Opus 4.8	Web (claude.ai)	51	32	79	48	10	80	300
Claude Opus 4.8	Claude Code	3	87	51	39	33	108	321
Claude Opus 4.8	AutoFyn	16	109	492	110	33	137	897
GPT-5.6 Sol	Web (chatgpt.com)	3	23	13	5	3	15	62
GPT-5.6 Sol	Codex	3	13	14	7	4	10	51
GPT-5.6 Sol	AutoFyn	18	68	65	23	24	26	224
Claude Fable 5	Web (claude.ai)	3	65	157	11	8	47	291
Claude Fable 5	Claude Code	5	19	37	20	4	38	123
Claude Fable 5	AutoFyn	26	51	115	49	27	155	423
Claude Sonnet 5	Web (claude.ai)	5	17	24	64	43	17	170
Claude Sonnet 5	Claude Code	3	45	47	103	18	39	255
Claude Sonnet 5	AutoFyn	22	44	1200	82	22	1200	2570
GLM 5.2	Web (chat.z.ai)	9	11	24	19	13	34	110
GLM 5.2	AutoFyn	29	141	247	41	51	267	776

harnesses and $7/7$ on the four web problems it could submit, so even a strong model can be bounded by the harness rather than the mathematics.

5.2 How the Systems Solve

The successful attempts share a few patterns. Problem 2 is closed computationally in every $7/7$ cell, by an exact symbolic certificate rather than a synthetic argument. Problem 5 admits a short functional-equation argument and was solved faster than Problem 4 across every run that solved both. Problems 3 and 6 each turn on a single combinatorial or number-theoretic step, the tree-bipartition discrepancy of Section 5.1 and a minimal-counterexample descent, and every successful attempt passes through it.

The Problem 6 step is one the models had seen before, as IMO Shortlist 2013 N5. In the multi-agent harness, where the explorer surfaces past problems and solutions, the four closed-weight proofs adapt its minimal-counterexample descent as their load-bearing step, and the Claude Sonnet 5 write-up states the adaptation outright, “adapted from IMO Shortlist 2013 N5’s official Solution 1, with every step re-derived from the recursive definition.” Claude Opus 4.8, Claude Sonnet 5, and GLM 5.2 completed Problem 6 in no other harness.

We also note that the analog is a route, not a requirement. GPT-5.6 Sol’s Codex run proves Problem 6 without the descent, reaching the same conclusion through an intersecting family of prime supports with a divisibility criterion, and names no past problem.

5.3 A Graded Example

Next, we give one graded run in full to show how the completion-based standard is applied and what it costs a near-complete proof.

Claude Sonnet 5 worked Problem 3 through AutoFyn for twenty hours, obtained the correct value, and proved the full upper bound. The lower bound reduces the physical cutting game to an abstract model, and the reduction, which the report labels D/M completeness, is left unproved. That single missing step is load-bearing, so the run scores 0 despite a correct answer and a complete upper bound. The run’s own internal reviewer had marked the file solved; the audit did not accept that label and graded the proof on the missing reduction.

The standard is therefore not lenient toward near-misses, and it does not credit a correct answer or a sound partial structure. A proof scores full marks only when every load-bearing step is present, and any single such gap sends the score to 0 regardless of how much of the argument stands around it. This is the property that makes the totals comparable across models and harnesses.

5.4 What the Harness Adds

The harness moves the score most for the sub-frontier models. GPT-5.6 Sol varies by two points across its three harnesses and Claude Fable 5 scores $42/42$ through both agent harnesses, whereas Claude Opus 4.8 gains 14 points and Claude Sonnet 5 gains 21 moving from their provider harness to AutoFyn, and the open-weight GLM 5.2 gains 14 from the web to AutoFyn. The two systems that already complete the contest are nearly harness-independent; the sub-frontier models are where the multi-agent orchestration matters.

Its contribution is specific and identifiable. AutoFyn recovers Problem 2 for GPT-5.6 Sol and adds it for Opus 4.8, the problem whose certificate requires a long exact derivation that the harness's review loop retains and verifies (Section 5.1). It adds Problem 6 for Opus 4.8, Sonnet 5, and GLM 5.2, where the subagent search surfaced the past analog that the completed proofs adapt (Section 5.2), and it additionally recovers Problem 4 for Sonnet 5 and Problem 5 for GLM 5.2. In these cases the harness supplied verification or retrieval for a step the model could reach once it was checked or found.

It does not, however, supply a missing capability. Problem 3's load-bearing step had no past analog to retrieve and its lower bound admits no certificate to verify, and there the orchestration changes nothing; every sub-frontier model, GLM 5.2 included, scores 0 on Problem 3 in AutoFyn, and the strongest such attempt stalls after twenty hours (Section 5.1). The harness closes the distance between a reachable proof and a submitted one, but not the distance between a model and an argument it cannot find.

6 Limitations

- **Small matrix.** We evaluate five models over three harnesses on a single contest, filling 14 of the 15 cells; GLM 5.2 was run through the web and AutoFyn only. The matrix is therefore small, and the trends we report rest on a limited number of cells.
- **Single attempt.** We cover one olympiad in one year with a single attempt per cell ($n = 1$) and no averaging over seeds or repeated attempts, so a shared failure across models cannot yet be separated from a coincidence of single draws. In future we aim to run the same claims at $n = 3$ per cell, which would let us report per-cell variance and test whether the shared-failure pattern holds under repetition. As a first probe, a second AutoFyn run of Claude Sonnet 5 scored $28/42$ against the archived run's $35/42$ (shown as a dot in Figure 1), a spread we report but do not yet average over.
- **Scores are not official grades.** The reported scores are the authors' assessment, adopted from an independent LLM audit and reviewed by a panel of three past IMO medalists, one a gold medalist leading it, and they are not official IMO jury grades, though a model is itself part of the audit loop for other runs and never for its own.
- **Harnesses run at defaults.** Each harness was run at its own default settings, with only two uniform limits on top, a 20-hour per-problem budget and a five-attempt retry rule (see Section 3.3), so the totals reflect what each deployment produces out of the box rather than a controlled head-to-head. The axes are not held fixed across cells, and we draw no causal ranking of harnesses.
- **Harness set is not exhaustive.** We report the three harnesses we ran end-to-end under our own audit and archival protocol. Other public harnesses exist, including the open verification-and-refinement pipeline of Huang and Yang [2025]; running such a harness on the 2026 problems under this protocol is a natural extension we aim to do in future.
- **Freshness is a protocol claim.** Freshness is a claim about the declared environment and not an absolute guarantee that a problem was novel to a hosted model.

7 Conclusion

On a fresh olympiad, graded harshly and audited outside the loop that produced the proofs, two systems completed all six problems, the two sub-frontier Claude models reached $35/42$, and the open-weight GLM 5.2 reached $28/42$, each through the strongest harness we ran. The evaluation protocol and the archived trails are what let us report this rather than take it on self-report. They also show that a score is not a property of the model alone but of the model together with its harness, a dependence large enough that a single number is incomplete without it. For the sub-frontier models the dependence is worth a medal: the two Claude models move from bronze to gold as the harness changes, and GLM 5.2 from no medal to one point short of gold. The matrix of five models against three harnesses is complete but for one cell, every run's trail is archived for scrutiny, and AutoFyn is described in the companion technical report [Hasan et al., 2026].

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